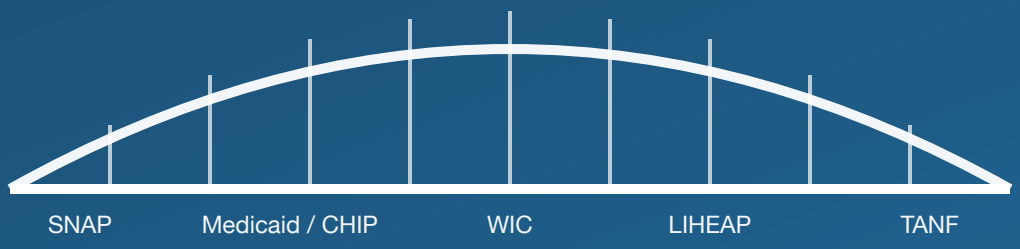
WHITE PAPER · PHASE 1 PROTOTYPE

OpenBridge

An open-source, privacy-first benefits navigator that helps families find and apply for the help they already qualify for, in one sitting.



What it is, what it is meant to accomplish, how it works, where it falls short today, and what it will take to get it into people's hands.

Version 1.0 · November 2026

Tyler Starling · StarlingTek, LLC

1. Executive Summary

Millions of people in the United States qualify for food, health, and energy help that they never get. The programs are there. The funding is there. What breaks down is the path in.

Think about what applying actually looks like. A parent sits at a library computer with a few hours before pickup. SNAP has one website. Medicaid has another. WIC means a phone call and a clinic visit. Each one asks for the same income details in a slightly different way, and none of them mention that the family probably qualifies for the others too. Plenty of people give up partway through, and who could blame them?

OpenBridge is an open-source web app built to make that afternoon easier. You answer one short set of plain-language questions about your household and income. OpenBridge checks those answers against several programs at once: SNAP, Medicaid and children's coverage, and WIC today, with LIHEAP, TANF, and others to follow. For each one it tells you where you likely stand and why. Then it fills in what it can of the application paperwork and points you to real people when your situation is complicated.

It does this **without an account and without sending your information anywhere.** Every calculation happens inside the browser, and nothing is saved unless you choose to download an encrypted copy. We built it this way on purpose. The people who need these programs most, including immigrant families and anyone on a shared public computer, are often the people with the most reason to be careful about what they type into a website.

5	3	2	0
programs modeled in the engine	pilot states: Ohio, Texas, Washington	languages: English and Spanish	bytes of personal data sent to a server

> **WHERE THINGS STAND**
>
> OpenBridge is a working prototype, but it isn't ready for the public yet. The code builds and the automated tests pass. Nobody outside the project has checked the eligibility rules, though, and that review matters more than anything else here. Legal review hasn't started either, and there's no live site. Until those steps are done, please treat any result as a rough guide and nothing more. Sections 7 and 8 go into the details.

2. The Problem

We tend to talk about "the safety net" as if it were one thing. It isn't. It's dozens of programs run by different federal agencies and administered by fifty states, and each one draws its own lines. SNAP and Medicaid don't even agree on who counts as part of your household. A family that qualifies for one program usually qualifies for two or three more, but finding that out is left almost entirely to the family.

The numbers give a sense of how many people slip through:

Finding	Source
About 5.7 million people enrolled in SNAP or Medicaid are eligible for WIC but not enrolled	USDA
About 9 million older adults are eligible for SNAP but not enrolled	National Council on Aging
About 3.7 million children are eligible for Medicaid or CHIP but uninsured	Georgetown Center for Children and Families
About 28% of eligible workers never file for unemployment insurance	WorkRise / EPI

These figures come from different years and different methods, so read them as a sense of scale, not exact counts. We'll re-check each one against current agency data before using it in any public claim.

Why people fall through

Some of it is simple fragmentation: every program is its own application, often on its own website. Some of it is complexity. Rules change from state to state and from program to program, and the language on official sites assumes you already know the system.

A lot of it is fear. In a mixed-status family, a parent may worry that applying for a child's health coverage will draw attention to someone else in the home. That worry keeps eligible kids uninsured. Others are just wary of any website that asks how much money they make, and honestly, that's a reasonable instinct.

Then there are the practical barriers. Many people apply from a library computer or an older phone. Many read below a high-school level or are more comfortable in a language other than English. And sometimes one confusing "you do not qualify" message, even a wrong one, is enough to end the search for good.

Good screening tools already exist, and we don't want to pretend otherwise. But many cover a single state or a single program, ask you to create an account, keep your data on their servers, or give an answer without showing how they got there. OpenBridge is meant to be open, portable, and easy for any community organization to run on its own.

3. What OpenBridge Does

A session with OpenBridge is meant to fit inside a single 30-minute visit to a library or clinic. It moves through five steps:

1. Welcome	2. Household	3. Income	4. Results	5. Act
Pick a language and mode, and see exactly what stays on the device.	State, county, who lives together, ages, pregnancy, disability.	Each source, how much, and how often, in whatever format is easiest.	One result per program, with the reasons behind it.	Download forms, save progress, find help, or wipe everything.

What it can do

You only answer once. The same household and income answers feed every program, so nobody types their paycheck in five times.

It shows its work. Every result comes with the reasons behind it, in plain words. Instead of a bare "eligible," you might see that your monthly income of $2,100 is under the $2,830 limit for a household your size. The English text is written to read at about a 6th to 8th grade level, and we check that automatically.

It looks at each person, not just the household. For Medicaid and WIC, results are shown person by person. That matters more than it sounds. In many states the children qualify even when a parent doesn't, and a single household-wide "no" would hide that.

It leans toward encouragement, but it doesn't overpromise. There are five possible outcomes: eligible, likely eligible, possibly eligible, ineligible, and unknown. OpenBridge won't give a flat "no" when someone is close to a limit, when an answer is missing, or when a rule is something it can't actually check. Immigration status on its own never leads to "ineligible."

It gets you started on the paperwork. You can print a summary of everything you entered along with each result. In Ohio you also get a pre-filled SNAP and Medicaid application. Spanish speakers get a companion guide that walks through filling out the official form, which is only available in English.

It doesn't leave you at a dead end. If nothing looks likely, OpenBridge explains why without making anyone feel turned away, then points to legal aid, 211, and local help.

It's safe on a shared computer. A "Clear my data" button is always on screen, the app wipes itself after a period of inactivity, and nothing is written to the browser's storage.

It works for helpers too. In navigator mode, a caseworker or volunteer can screen several families in a row, locking the screen and moving to the next client by hand instead of relying on the timer.

What it isn't

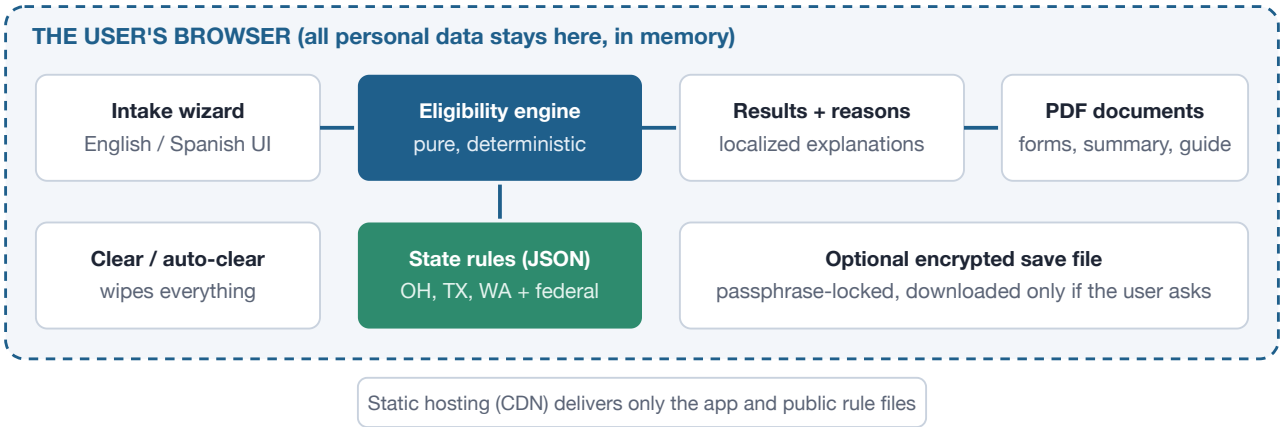
OpenBridge doesn't submit applications, connect to government systems, make eligibility decisions, give legal advice, or keep anyone's records. Its job is narrower than that. It helps people understand where they likely stand and gets them to the official application with as much of the groundwork done as possible. A person always reviews and submits the application themselves.

4. Who It Serves

User	Who they are	What OpenBridge gives them
Benefit seekers	Individuals and families, including people with low literacy or limited tech experience, non-English speakers, and mixed-status households on shared devices	One private, plain-language session that ends with a clear next step
Community navigators	Caseworkers and volunteers at food banks, libraries, health centers, and 211 lines	Quicker screening across programs for many clients, with no new accounts or data to manage
Community organizations	Groups that want to offer screening on their own site	A free app they can host themselves or customize with a small settings file
Contributors	Developers, policy experts, translators	A setup where adding a state is mostly data and test cases, not new code

5. How It Works

OpenBridge is a Progressive Web App, which is a website that can be installed like an app and keeps working offline after your first visit. In this first phase there's no backend server at all.



An engine you can check

At the center of OpenBridge is a small eligibility engine. You hand it a household and a state's rules, and it hands back a result for each program. It doesn't go online, doesn't remember anything between runs, and doesn't involve any guesswork. The same answers always produce the same result. That sounds like a minor technical point, but it's what lets us test the engine against sample households whose correct outcomes have been worked out by hand, and catch mistakes before anyone relies on them.

The engine also handles the details that simpler screeners tend to get wrong. Each program builds its own idea of the household from the same list of people. Pay can be entered hourly, weekly, every two weeks, or yearly, and it all gets converted to a monthly figure. Money is tracked to the cent so rounding doesn't push anyone over a line, and Alaska and Hawaii get their own poverty-level tables, as federal rules require.

States as data, not code

Each state's rules live in their own data file, which we call a **state adapter**. It holds the income limits, whether the state expanded Medicaid, its SNAP waivers, and links to its applications. Every number in it has to cite an official source. To add a new state, someone writes that file and a set of sample households, and an expert checks both. The engine itself doesn't change. That's what makes covering all fifty states a realistic goal for a community project rather than a fantasy.

Privacy built into the design

A privacy policy is a promise. OpenBridge's privacy comes from how it's built. All the math runs on your device, so your information is never sent anywhere. While you work, your answers live only in the app's short-term memory, not in browser storage or cookies, and every screen that asks a question says so in plain words. If you want to come back later, you can download a save file locked with a passphrase you choose, using standard encryption (PBKDF2 and AES-GCM). We never see that passphrase, so there's no way to recover a forgotten one. In this first phase the app collects no analytics and no error reports at all.

Made for the devices people actually have

OpenBridge is built to run on older library browsers and cheap phones. It works offline after the first load, and the initial download is about 119 KB compressed, comfortably under our 250 KB limit. It's designed to meet the WCAG 2.2 AA accessibility standard, and it's licensed under AGPL-3.0, so anyone who improves it has to share those improvements back.

6. Current Status

We've worked through the whole Phase 1 plan. As of this writing, the code builds cleanly and these checks pass:

78	8	3	6.6
automated unit tests passing	sample test households passing	state rule files passing validation	reading grade level of English text (target 8 or lower)

Passing tests only tells you the code does what it was written to do. It doesn't tell you the rules themselves are right, and that's why the table below has a few rows that say "not started."

Area	State	Notes
Eligibility engine (SNAP, Medicaid, WIC, LIHEAP, TANF)	Built	Tested against hand-worked limits and edge cases
Ohio, Texas, Washington rules	Draft	Example 2024 figures entered by the development team; expert review still to come
Intake, results, program detail screens	Built	English and Spanish
Privacy controls (clear, auto-clear, privacy banner)	Built	Data held in memory only
Encrypted save and resume	Built	Save, reopen, and wrong-passphrase cases tested
Documents (summary, Ohio prefill, companion guide)	Built (basic)	The Ohio prefill lists the right fields but doesn't yet look like the official form
Referral directory	Seeded	5 entries across the three pilot states
Offline support	Built	The app and its rules are cached on the device
Hosting setup	Written, not deployed	Cloud configuration is ready; there's no live site yet
Legal review	Not started	Review package is prepared
Expert review of rules	Not started	Reviewers in Franklin County, Ohio have been identified

7. Limitations and Known Gaps

We've kept Phase 1 narrow on purpose. Some of its limits are choices we made to protect the people using it. Others are simply work we haven't gotten to yet. We'd rather list all of them plainly than have anyone mistake a prototype for a finished tool.

7.1 Limits we chose

It screens; it doesn't decide. Only the agency can say whether someone qualifies. Since OpenBridge can't verify anything a person tells it, most results come back as "likely" or "possibly," and that's intentional.

It doesn't submit anything. States rarely let outside tools file through their systems, so people still need to copy their information into the official portal or onto a paper form. OpenBridge makes that easier. It can't skip that step for them.

There are no accounts and no cloud history. That protects privacy, but there's a cost. Lose the save file or forget the passphrase, and you start over. Nobody can recover it for you.

We can't see how it's used. With no analytics, we don't know how many people use OpenBridge or where they get stuck. For now we'll have to rely on what partner organizations tell us.

Official forms are mostly English-only. We can't change that. Spanish speakers get a companion guide to help them through the English form, but not a translated application.

7.2 Programs and rules not yet covered

- **Medicaid is income-based only.** Coverage built around age, disability, high medical bills, or "spend-down" isn't evaluated. When someone's answers suggest one of those might apply, the app says so and suggests talking to a navigator.
- **CHIP is approximated.** It's handled through Medicaid's children's income limits, not as a separate program with its own premiums and rules.
- **WIC tops out at "possibly eligible."** Part of qualifying is a nutrition check that only happens in person at a WIC clinic.
- **Some SNAP rules aren't calculated.** The work requirement for adults without dependents is flagged but not worked out. Vehicle value isn't assessed, and self-employment costs can't be itemized.
- **LIHEAP and TANF are early.** Both are in the engine, but full support is planned for Phase 2. TANF varies so much from state to state that screening for it will always be rough.
- **Many programs aren't included yet,** among them Unemployment Insurance, SSI and SSDI, housing assistance, child care subsidies, the Earned Income Tax Credit, school meals, and county or city programs.

7.3 Accuracy

This is the gap we worry about most. All three state rule files are still drafts. The app does show a "rules still being reviewed" notice, but until experts sign off, the results aren't reliable enough for the public. The numbers are also from 2024. Federal poverty levels change every January and SNAP figures every October, so everything needs a refresh before launch.

Medicaid households are another weak spot. Officially, they're defined by who files taxes and who claims whom. OpenBridge estimates this from relationships and ages, plus two optional tax questions, and gives a more cautious answer when the estimate could change the outcome. Even so, households with shared custody, adult dependents, or a parent outside the home claiming a child may be judged wrong.

We also need far more test cases. Eight sample households is a start, but a release people can trust needs dozens per state, especially near the limits and in unusual situations. And no matter how good the engine gets, it's only as accurate as the answers people give it. Income questions are where we expect most real-world mistakes.

7.4 Product and operations

- **Only three states are covered.** Anyone outside Ohio, Texas, or Washington will see "unknown" for now.
- **The Ohio pre-filled application isn't the official form yet.** It has the right fields in a clean list. Making it match the real Ohio Benefits layout means getting the current form and mapping each field to it.
- **The referral directory is tiny.** Five entries can't serve three states. It needs county-level listings and someone to keep them current.
- **Accessibility has only been checked by software.** The automated checks pass. Real testing with screen readers and with people who have disabilities hasn't happened yet.
- **Nobody outside the project has used it.** We haven't put the intake in front of actual benefit seekers or navigators. The wording, the length, and the optional tax questions all need real feedback.
- **It isn't live or public yet.** There's no hosted site, and the code hasn't been published or put under shared version control.
- **The navigator tools are basic.** A fuller dashboard for organizations is planned for Phase 2.

8. Hurdles to Overcome

These are the obstacles between the prototype and something people can rely on. They're ordered roughly by how much each one blocks a launch.

BLOCKER Getting the rules right, and keeping them right

A wrong answer can cost a family their food assistance or their kids' health coverage, or send them through an application that ends in a denial. Each state's rules need review by people who run or advocate for these programs. We're starting with social work and epidemiology reviewers in Franklin County, Ohio. We haven't found reviewers for Texas or Washington yet. And the work doesn't stop at launch. Every January and October the numbers change, and each update has to be checked and released again. Someone needs to own that job for the long run.

BLOCKER Legal review

OpenBridge has to stay clearly on the side of helping people fill out forms, and well away from anything that looks like legal advice. That line is known as the unauthorized practice of law, and it's drawn differently in different states. A lawyer needs to review the Terms of Use, the disclaimers, the referral directory and its no-endorsement policy, the privacy statements, and the choice of license. We won't launch until that review signs off.

BLOCKER Earning trust where it matters most

The design keeps people's data private, but they have to believe that. Mixed-status families have every reason to be careful. Trust will come from the local organizations they already rely on, from clear explanations of what happens to their information, and ideally from an independent review of our privacy claims. Without it, the people who need OpenBridge most may never try it.

MAJOR Growing to fifty states

The design makes it possible. It doesn't make it easy. Every state still has to be researched, written up, tested, checked, and then maintained. That takes a community of contributors, clear guidelines, and a review process that holds up as the number of states grows. Official form templates are a separate job on top of that, one state and one program at a time.

MAJOR Getting organizations to use it

Navigators are stretched thin and already have their tools. For OpenBridge to earn a place, it has to be faster than what they do today, fit how they already work, and come with some training. Our first pilot partnerships, starting in Franklin County, are how we'll find out what actually helps.

MAJOR Showing impact without tracking anyone

Funders and partners will reasonably want proof that OpenBridge gets more people enrolled. Since Phase 1 collects nothing, that evidence has to come from partner reports. We plan to add a privacy-safe

way to count outcomes in aggregate, but we haven't designed it yet.

MAJOR Keeping it going

Hosting a site like this costs very little. Keeping the rules current, paying for expert review, maintaining translations, and supporting partners costs a lot more. OpenBridge needs a long-term home, whether that's a nonprofit, a civic-tech group, or a coalition, along with clear rules about who approves changes and steady funding for review cycles.

MINOR Forms and portals change

States revise their forms and websites without much warning. Our documents will have to keep up, and some agencies may be reluctant to accept anything that doesn't look exactly like their own paperwork.

MINOR More languages

English and Spanish reach a lot of people, but not everyone. Each new language needs skilled translators who can keep up with rule changes and keep the wording simple.

MINOR Working alongside other tools

Other screeners and civic-tech projects are already doing good work. We see OpenBridge as shared, open infrastructure that can partner with them or power their tools, not as a competitor.

9. Roadmap

Phase	Focus	Key deliverables
Phase 1 Core and pilot	Prove the idea works in three states	SNAP, Medicaid, and WIC screening; OH, TX, WA; English and Spanish; offline app; Ohio forms; save and resume on the device; referrals; expert sign-off; legal review; public hosting
Phase 2 More programs and states	Broader coverage and better tools for navigators	Full LIHEAP and TANF; about 10 states; more official form templates; organization dashboard; custom branding; training materials; more languages
Phase 3 Tracking and integration	Optional cloud features for organizations	Opt-in encrypted cloud sessions; staff accounts for organizations; Unemployment Insurance; connections to case-management tools; about 25 states; public open-source release; privacy-safe impact counts

Phase 4 Scale and a lasting home	Nationwide coverage and long-term ownership	All states for the core programs; 211 integration; a governance model; a clear owner for the twice-yearly data updates

10. Risks and Mitigations

Risk	How we're handling it
Wrong eligibility math harms people	Sample households tested on every build; expert sign-off before a state is marked verified; cautious outcomes; a visible "draft rules" notice
State rules change often	Every number cites its source and date; scheduled January and October updates with fresh review; an "out of date" warning in the app
Agencies reject generated documents	Match official layouts; always include guidance for entering the same information in the official portal
Crossing into legal advice, or liability from referrals	Stay within form help; Terms of Use; a directory that leads with legal aid and takes no paid placements; legal review before launch
Sensitive data left behind on shared computers	Data held in memory only, a one-click clear button, automatic clearing, and encrypted saving only
Personal data leaking through analytics or error reports	None of either in Phase 1; any future reporting limited to a short, approved list of non-personal details
Low trust in immigrant communities	No accounts; status never rules anyone out on its own; plain explanations of data handling; outreach through trusted local partners
Maintenance grows with every state and program	Rules kept as data; modular design; contributor guidelines; a governance model in later phases

11. How to Help

OpenBridge needs more than code right now. Here's where help would make the biggest difference:

Expertise

- **Benefits and policy experts** to review state rules and write test households, especially for Texas and Washington
- **Lawyers** who know benefits access and the rules around giving legal advice

Partnership

- **Community organizations** willing to pilot it with their navigators and tell us honestly what works
- **People who use benefits** willing to try the questions and point out what's confusing

- **Accessibility testers**, including people who use screen readers
- **Translators** to review the Spanish and add new languages
- **Civic-tech groups** interested in shared tools or in giving the project a long-term home
- **Funders** who can support expert review and data upkeep

WHAT COMES NEXT

1. Finish expert review of the Ohio rules with the Franklin County reviewers, then move on to Texas and Washington.
2. Update all reference numbers to current federal and state figures.
3. Get legal review of the Terms of Use, the disclaimers, and the referral directory.
4. Add many more sample test households, and grow the referral directory to county level.
5. Build the Ohio form templates from the current official Ohio Benefits applications.
6. Run test sessions with benefit seekers and navigators, along with hands-on accessibility testing.
7. Publish the code and launch a pilot site for partner organizations.

12. Conclusion

The distance between the people who qualify for help and the people who actually receive it isn't mainly about money. It's about finding your way through the system, and that's a problem software can make better. OpenBridge tries to do that carefully. Ask once. Check every program. Explain every answer. Never turn away someone who might qualify, and never collect more than the job requires.

The technical groundwork is done, and it's been tested. What's left is the human work that makes it trustworthy: experts checking the rules, lawyers checking the boundaries, real people trying it out, and a community willing to keep it accurate year after year. If we get that right, any library, food bank, or clinic will be able to offer OpenBridge, free, to the families who walk through its doors.

OpenBridge is a screening and form-help tool. It does not decide eligibility, give legal advice, or submit applications. The program rules and figures in this paper are based on draft reference data and must be checked against current agency sources before public use.